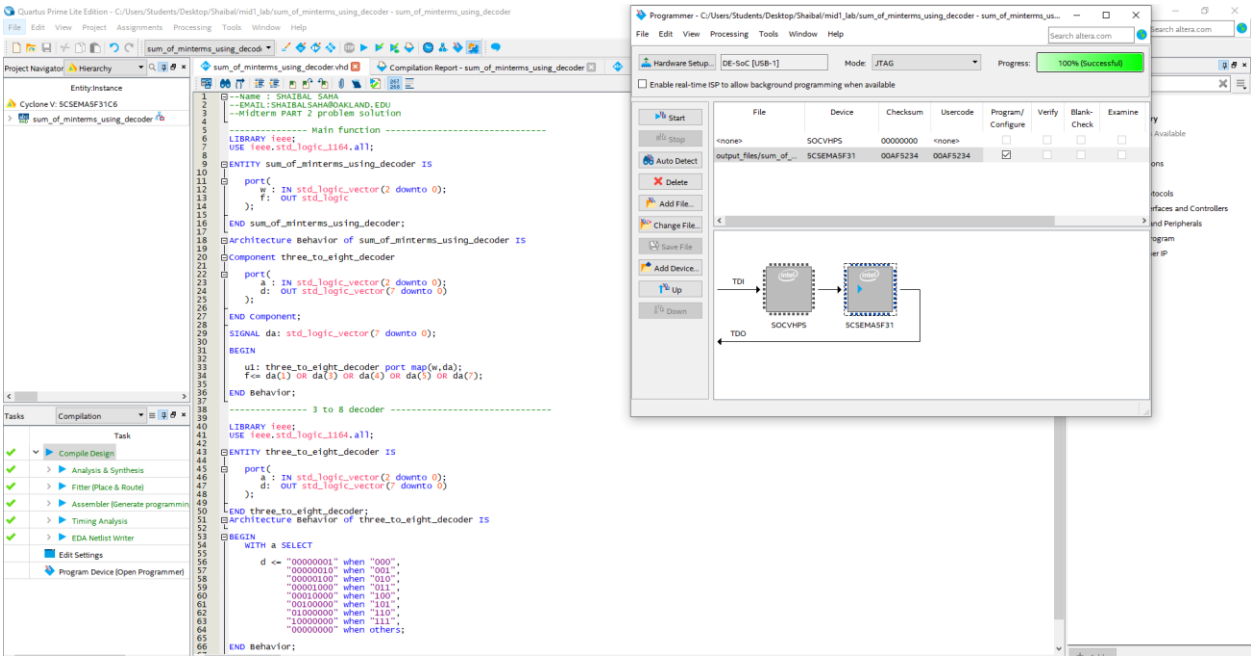
This problem is the part of midterm 1. Here in this report ,I will show all the screenshot of the code as well as waveform from test branch outputs and screenshot on the output in the DE-soc1 board.

The problem is to design a sum of minterms **f(w1,w2,w3) = ∑(1,3,4,5,7)** . I need to implement this with 3 to 8 decoder and one OR gate. The below code is for the 3 to 8 decoder and the function that implement the or gate:



```
--Name : SHAIBAL SAHA
--EMAIL:SHAIBALSAHA@OAKLAND.EDU
--Midterm PART 2 problem solution

------------- Main function ------------------------------
LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY sum_of_minterms_using_decoder IS

   port(
      w : IN std_logic_vector(2 downto 0);
      f: OUT std_logic
   );

END sum_of_minterms_using_decoder;

Architecture Behavior of sum_of_minterms_using_decoder IS

Component three_to_eight_decoder

   port(
      a : IN std_logic_vector(2 downto 0);
      d: OUT std_logic_vector(7 downto 0)
   );

END Component;

SIGNAL da: std_logic_vector(7 downto 0);

BEGIN

   u1: three_to_eight_decoder port map(w,da);
   f<= da(1) OR da(3) OR da(4) OR da(5) OR da(7);

END Behavior;

------------- 3 to 8 decoder ------------------------------

LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY three_to_eight_decoder IS

   port(
      a : IN std_logic_vector(2 downto 0);
      d: OUT std_logic_vector(7 downto 0)
   );

END three_to_eight_decoder;
Architecture Behavior of three_to_eight_decoder IS

BEGIN
    WITH a SELECT

        d <= "00000001" when "000",
             "00000010" when "001",
             "00000100" when "010",
             "00001000" when "011",
             "00010000" when "100",
             "00100000" when "101",
             "01000000" when "110",
             "10000000" when "111",
             "00000000" when others;

END Behavior;
```

Figure: 3 to 8 decoder and the function that implement or gate

	tatu	From	To	Assignment Name	Value	Enabled	Entity	Comment	Tag
1	✓		w[0]	Location	PIN_AB12	Yes			
2	✓		w[1]	Location	PIN_AC12	Yes			
3	✓		w[2]	Location	PIN_AF9	Yes			
4	?		SW[3]	Location	PIN_AF10	Yes			
5	?		SW[4]	Location	PIN_AD11	Yes			
6	?		SW[5]	Location	PIN_AD12	Yes			
7	?		SW[6]	Location	PIN_AE11	Yes			
8	?		SW[7]	Location	PIN_AC9	Yes			
9	?		SW[8]	Location	PIN_AD10	Yes			
10	?		SW[9]	Location	PIN_AE12	Yes			
11	✓		f	Location	PIN_V16	Yes			
12	?		LEDR[1]	Location	PIN_W16	Yes			
13	?		LEDR[2]	Location	PIN_V17	Yes			
14	?		LEDR[3]	Location	PIN_V18	Yes			
15	?		LEDR[4]	Location	PIN_W17	Yes			
16	?		LEDR[5]	Location	PIN_W19	Yes			
17	?		LEDR[6]	Location	PIN_Y19	Yes			
18	?		LEDR[7]	Location	PIN_W20	Yes			
19	?		LEDR[8]	Location	PIN_W21	Yes			
20	?		LEDR[9]	Location	PIN_Y21	Yes			
21		<<new>>	<<new>>	<<new>>					

Figure: Pin assignment for desoc1 board output

The next screenshot is the code for the test inputs and outputs .

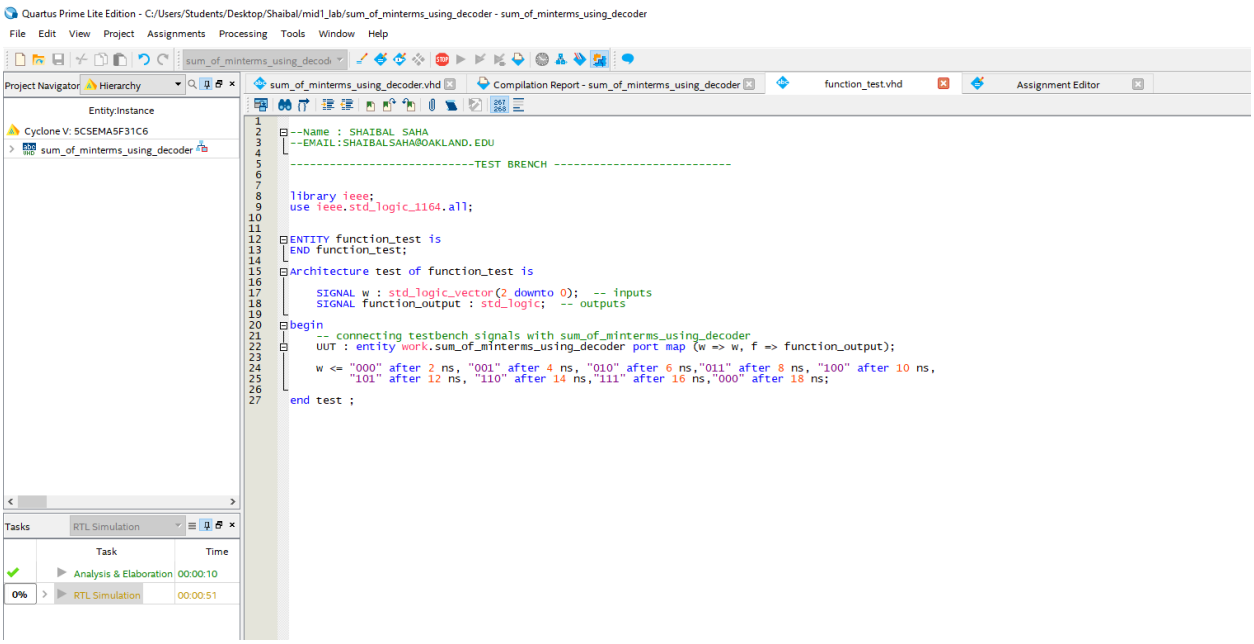


Figure: Test branch code

Output: I simulate in the modelsim altera as well as testing with desoc1 board. Here I attach both the screenshot and picture taken from de-soc1 board some of the inputs.

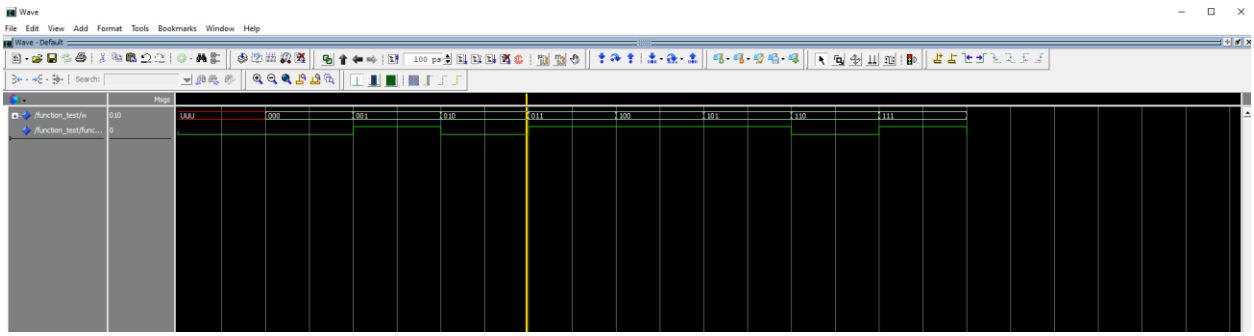
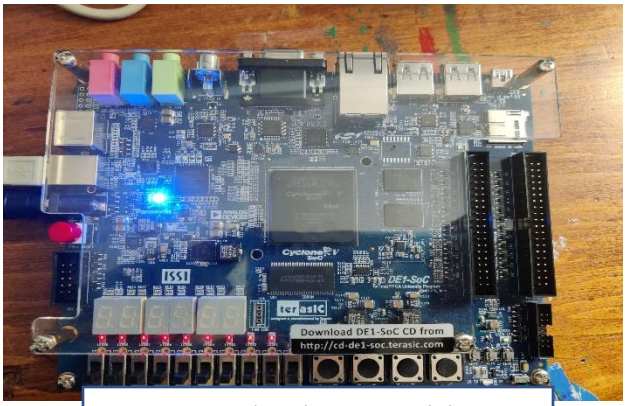
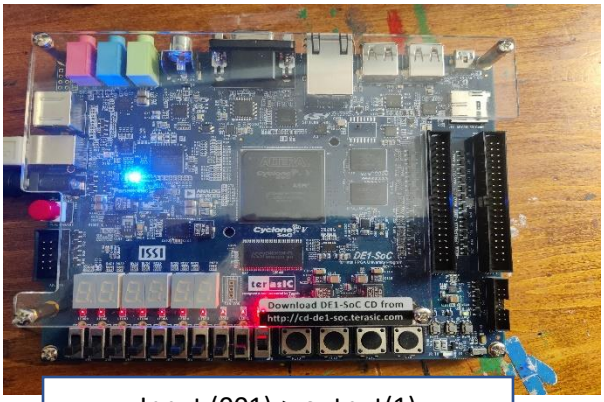


Figure: Waveform of the all possible input



Input (000)-> output(0)



Input (001)-> output(1)

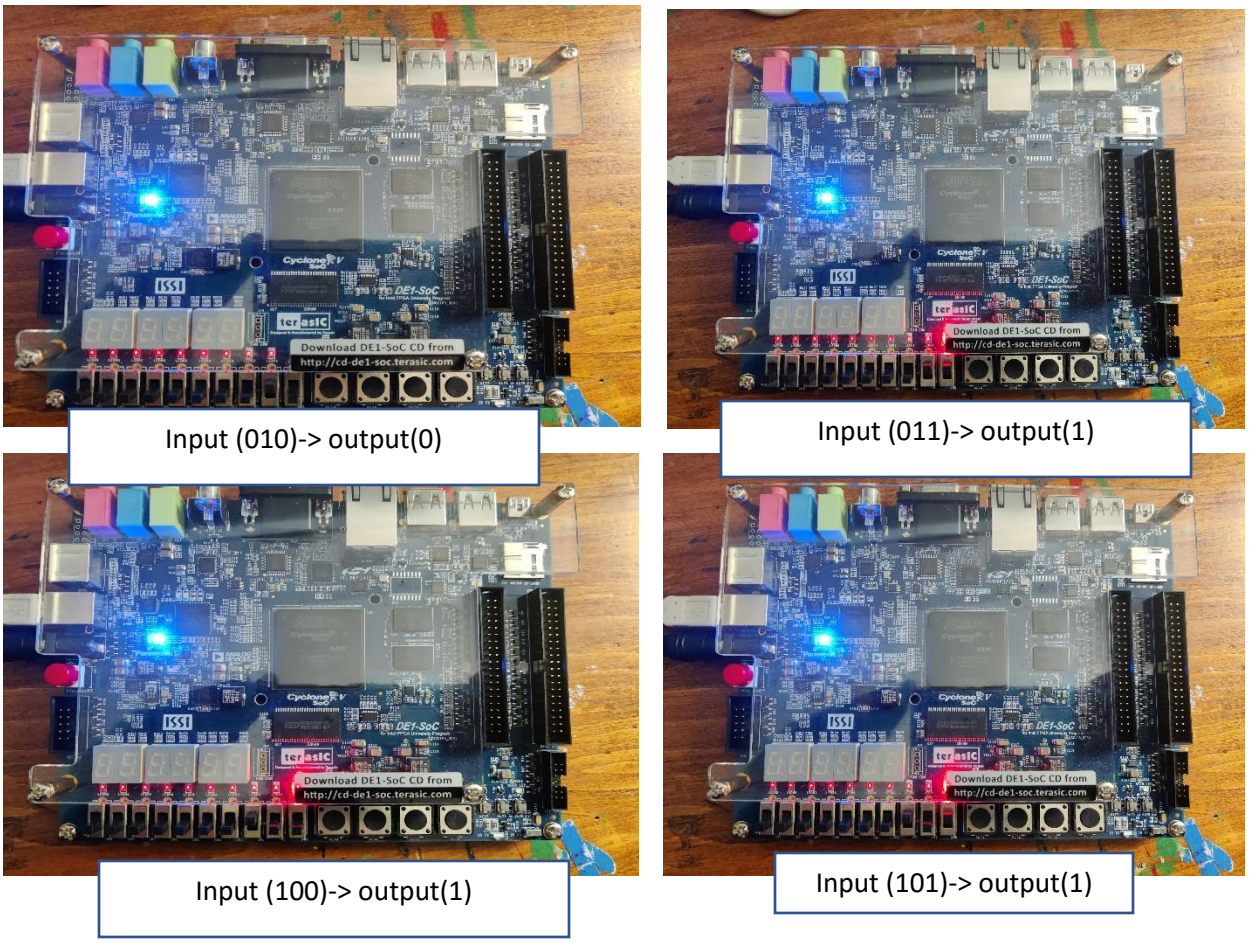


Figure: Output in DE-SOC1 board

Note: SOURCE CODE IS IN THE **mid1_lab** folder.